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Students-Generative AI interaction patterns and its impact on academic writing

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Abstract

Considering both the transformative opportunities and challenges presented by generative AI (GenAI) in academic writing, effectively integrating GenAI into the academic setting becomes a significant need requiring prioritization. Yet, there is limited understanding regarding the nature of interactions between different types of students, what behavioral patterns students exhibit during a student-GenAI interaction (SAI) on a given task, and how these different SAI patterns relate to the actual writing task performance. This study, therefore, aimed to identify SAI patterns of academic writing tasks depending on students' level of AI literacy and examine the differences in academic writing performance between the identified SAI patterns. Drawing from the combination of three data sources, including think-aloud protocols, screen-recordings, and chat histories between 36 Chinese graduate students and a GenAI writing system, epistemic network analysis (ENA) was used to reveal the distinctive SAI patterns of students with different levels of AI literacy. The study found that students with a high level of AI literacy exhibited a collaborative approach to SAI, actively accepting GenAI's suggestions and engaging GenAI in meta-cognitive-related activities such as planning, whereas students with a low level of AI literacy demonstrated much less interaction with GenAI in completing their writing tasks, instead choosing to ideate and evaluate independently. In addition, the Wilcoxon rank-sum (Mann-Whitney U) test was conducted to assess the writing task performance of the two AI literacy groups. Findings revealed statistical differences in all evaluation rubrics (content, structure/organization, expression). This study offers implications for the design and implementation of GenAI agents in writing tasks and the pedagogy of GenAI-assisted instruction.

Keywords GenAI-assisted writing · AI in education · AI literacy · student-AI interaction · Interaction patterns · Epistemic network analysis

Extended author information available on the last page of the article

Introduction

Integrating generative AI (GenAI) into academic writing has generated substantial hype within the academic and language education community. Academic writing is a complex and meticulous learning process necessitating proficiency in multiple areas such as research skills, structured and coherent argumentation, and clear expression; however, students increasingly utilize AI-powered tools to enhance their writing processes and performance. For instance, GenAI-powered writing tools offer automated feedback on various aspects of writing, including organization, vocabulary choices, grammatical and lexical errors, and proofreading to improve the overall writing quality and structure (Malik et al., 2023). Learners can expedite their writing development by utilizing GenAI tools to generate initial drafts by building upon the learner's initial ideas and providing valuable suggestions for improving the existing text (Črček & Patekar, 2023). Furthermore, AI-powered search engines and databases facilitate the identification of relevant research articles and can offer personalized recommendations based on user preferences and previous search behavior (Behrooz et al., 2023; Chichekian & Benteux, 2022). Additionally, AI-driven citation and reference management software aids students in organizing and formatting their reference lists accurately (Zhang, 2012), ensuring compliance with various citation styles (Sharifi et al., 2021).

While the potential benefits of GenAI tools in academic writing are undeniable, there is a noteworthy dearth of knowledge regarding the dynamics of students-GenAI Interaction (SAI), the process of interacting between students and AI in learning task operation (Kim et al., 2024). This includes the nature of interactions between different types of students and GenAI agents, how they engage with GenAI agents during academic writing tasks, the variations in their task-completion flows, the nature of their relationship during a given task, and the impact of these different SAIs on task performance. Considering that prior studies in SAI demonstrate that students anthropomorphized AI agents as peers, tutors, and teaching assistants (Engwall & Lopes, 2020; Kim & Cho, 2023), students and GenAI may form a learning community wherein students perceive and experience cognitive, social, pedagogical, and technological influences during SAI and apply them to the task operation and learning experience (Jeon, 2022; Kim et al., 2024). Along with this, the literature on SAI in learning tasks has indicated that learners exhibit different learning patterns and performance depending on individual characteristics, and thus individualized support and intervention should be designed assuming learners' heterogeneity (Wang et al., 2023a, b; Kim et al., 2024; Kim & Lee, 2023). As such, it is imperative to examine how students interact differently with GenAI on academic writing tasks to enhance our knowledge about the nature of learning that occurs in GenAI-assisted writing and inform future educational AI design and GenAI-assisted instructional design to meet the individual needs of students with diverse SAI patterns. This study, therefore, aims to identify SAI patterns of academic writing tasks and examine the differences in academic writing performance between the identified SAI patterns. In doing so, this study can unpack the SAI process in writing tasks and provide insights into the design and implementation of GenAI agents in writing tasks and the pedagogy of GenAI-assisted instruction.

Literature review

AI-assisted academic writing

Academic writing actively engages students in learning, providing them with different ways of constructing knowledge (Hyland, 2014). It is a complex task necessitating coordinating various cognitive skills to orchestrate, including managing memory resources, problem-solving, and goal setting (Allen & McNamara, 2017). Further, it requires students to demonstrate and apply critical thinking skills and domain knowledge while successfully communicating their perspective on a given topic to inform readers of their findings, conclusions, or recommendations (Swales & Feak, 2012).

Already challenging for students writing in their primary language, English as a Foreign Language (EFL) students face a host of additional challenges throughout the academic writing process, such as accurately communicating their thoughts in English, coherently crafting an argument, conducting translation activities, or revising their work (Huang et al., 2023; Malik et al., 2023). These issues are compounded by educators' limitations regarding a lack of time to personalize instruction for these students or offer timely feedback, reducing their opportunities to proactively improve their writing (Aldabbus & Almansouri, 2022; Gupta et al., 2022).

GenAI technologies have revolutionized the educational landscape by providing students with an always-available, interactive platform for writing enhancement that offers comprehensive support across multiple dimensions of academic writing (Nguyen et al., 2024). These tools enhance writing activities (WA) through ideation and drafting support, allowing students to brainstorm ideas and refine their writing through iterative AI interactions (Malik et al., 2023; Nguyen et al., 2024). They also support meta-cognitive activities (MC) by facilitating planning, orchestrating writing processes, and enabling continuous evaluation of work (Wang et al., 2023a, b; Molenaar, 2022). Socio-emotional (SE) factors influence student engagement and success in using AI effectively. Positive emotions can be fostered through immediate, constructive feedback which helps students manage potential frustrations in the writing process (Chan & Hu, 2023). These tools effectively function as on-demand writing tutors, supplementing traditional writing processes and addressing the limitations of delayed human feedback (Kim & Lee, 2023). For example, Grammarly has been shown to improve writing task performance by enhancing student grammar and writing quality (Malik et al., 2023). Given that such feedback is immediate, students also appreciate that it gives them time to incorporate improvements or suggestions before finalizing their work (Malik et al., 2023).

Nevertheless, not all students can realize these benefits on their own due to barriers to using GenAI effectively, such as a lack of guidance from educators on how to use the technology in an education setting, an inability to effectively prompt AI to get appropriate results, or weak critical thinking skills that interfere with their ability to effectively assess or result in over-reliance on AI-generated content (Črček & Patekar, 2023; Malik et al., 2023; Singh et al., 2023). For example, a study examining interaction patterns between doctoral students and AI-based chatbots found that students who iteratively refine a prompt until it produces relevant, accurate, and desired results experience a more productive writing session. Further, critically analyzing

generated content and integrating the chatbot's feedback and content into their writing resulted in improved performance. Additionally, students' cognitive load of the writing process was reduced as students collaborated with the chatbot on cognitively demanding tasks such as planning or editing. This enables students to focus on and devote more time to creative and complex tasks that AI cannot replicate (Nguyen et al., 2024). In contrast, some students merely used AI in a linear fashion where they typically copied generated content into their essays seemingly without assessing it, and, as a result, had lower performance outcomes on their final essay (Nguyen et al., 2024). This highlights that it is not through the mere use of AI that students experience genuine improvement in their writing process or outcomes, but instead it is through high-quality interaction with AI that students can optimize their writing experience and performance.

Differences in student-AI interaction (SAI) depending on students' characteristics

Integrating AI into educational settings has sparked considerable interest in understanding how students interact with this cutting-edge technology (Chango et al., 2021; Hew et al., 2023). Previous research in human-computer interaction has established that effective collaboration between humans and machines involves complex activities, including communication, joint action, and the ability to adapt to human-aware execution (Lemaignan et al., 2017). However, despite the growing use of AI in education, little is known about how students engage with AI to produce creative outputs and how student characteristics influence this interaction (Kim et al., 2024). Students' characteristics play a pivotal role in determining the effectiveness of their interaction with technology. For instance, students' attitudes toward technology are directly related to their perceived usefulness and ease of use of the technology (Li et al., 2021; Venkatesh et al., 2003). Furthermore, students' domain-specific skills significantly impact their experiences and interactions with technology (Li et al., 2022), which suggests that students' problem-solving processes and experiences with AI cannot be dissociated from their skills and the content domain of the technology (Kim & Lee, 2023).

The concept of literacy, initially associated with language proficiency, has evolved to encompass various skill sets required across different disciplines. In the educational context, the advent of AI has underscored the need for enhanced AI literacy among teachers and students (Walter, 2024). AI literacy entails a critical understanding of AI's black-box algorithms, effective interaction and collaboration with AI, and the ability to utilize AI as a tool diversely (Lee et al., 2023). A prerequisite for AI literacy is digital literacy, which enables individuals to use computational devices effectively. The confidence level in utilizing these devices can significantly influence how information systems incorporating AI algorithms are perceived and used (Bond et al., 2024; Delcker et al., 2024). Given this backdrop, it becomes imperative to investigate student-AI interaction (SAI) differences based on students' characteristics, particularly their AI literacy levels. More importantly, it is crucial to understand how students with varying AI literacy levels engage with AI and whether these differences impact their learning outcomes, such as academic writing.

Studies found that students with a higher level of AI literacy tend to exhibit more effective utilization of AI tools for academic writing tasks. For instance, students proficient in analyzing and evaluating AI content demonstrate a remarkable ability to discern the relevance and accuracy of information retrieved from AI-powered writing assistants (Chango et al., 2021; Walter, 2024). Moreover, these students often display more positive emotions towards AI, recognizing its potential as a collaborative partner rather than a mere tool (Hew et al., 2023; Wang et al., 2023a, b). Their enhanced understanding of AI allows them to leverage its capabilities more efficiently, leading to improved writing outcomes in terms of both quality and efficiency. Conversely, students with lower AI literacy may struggle to fully utilize AI's potential, often viewing it as a foreign or threatening entity (Delcker et al., 2024; Hall, 2024). This disparity highlights the need for enhanced AI education and training, particularly in integrating AI into writing tasks, to empower all students to effectively engage with this transformative technology.

Effects of SAI on writing tasks

Research exploring the outcomes of student-AI engagement in writing endeavors has progressively emphasized the significance of students' individual traits, especially their AI literacy, on writing performance (Kim & Lee, 2023; Liu, 2019). Students' AI literacy impacts their writing endeavors, spanning content depth, linguistic expression, and compositional structure. For instance, students with higher levels of AI literacy can be more adept at amalgamating an array of viewpoints and insights into their writing (Ayanwale & Molefi, 2024). Their AI proficiency allows them to seamlessly integrate the suggestions and feedback offered by AI, thus augmenting their written works with a more extensive assortment of concepts and data (Cardon et al., 2023). However, this proficiency should not be viewed in isolation. While AI literacy is a crucial factor, it functions alongside and is potentially amplified by these pre-existing academic skills. Students with strong baseline abilities in writing and critical thinking can leverage AI to refine vocabulary and structure, incorporating synonyms and alternative expressions to create more diverse and engaging prose (Roe et al., 2023). This proficiency in seamlessly blending AI-suggested language into their writing elevates their compositions' overall readability and allure (Chan & Hu, 2023). Additionally, students with considerable AI proficiency have a deep understanding of how to arrange their thoughts effectively (Chan & Zhou, 2023). They can employ AI feedback to refine the structure of their essays, ensuring a logical sequence of arguments and a cohesive storyline. This structural mastery can contribute to more polished and professional outcomes (Hew et al., 2023).

Nonetheless, the complexities appear more accentuated for students demonstrating lower competencies in AI literacy. They may struggle to effectively incorporate AI input, limiting the comprehensiveness and breadth of their compositions. Such students might encounter challenges in embracing synonyms or substitutions suggested by AI, consequently impacting the richness and dynamism of their discourse (Cardon et al., 2023). Furthermore, they might not possess sufficient expertise to streamline the framework of their writings utilizing AI feedback, potentially undermining their manuscript's logical continuity and coherence (Nieminen et al., 2021). The reviewed

literature underscores the significant role of AI literacy in shaping students' writing outcomes, highlighting distinct differences in writing performance between students with varying levels of AI proficiency. Based on existing literature, we propose the following hypotheses to guide our research:

1. There are significant differences in the frequency distributions of student-generative AI interactions (SAI) between students with high AI literacy compared to those with low AI literacy.
2. The SAI patterns will differ significantly based on students' AI literacy levels.
3. Writing performance will differ significantly between students with high and low AI literacy.

Methodology

Research context and participants

This study was undertaken at a joint-ventured (Sino-British) research-led international university in Suzhou, mainland China. This university offers courses primarily in English, necessitating a solid command of the language from its students. To ensure this proficiency, students who are non-native English speakers must attain a specified score on standardized English tests such as the Test of English as a Foreign Language Internet-based Test (TOEFL iBT), or the International English Language Testing System (IELTS) before admission. The university's commitment to English proficiency extends beyond admission requirements and is reflected in its curriculum, requiring graduate students to engage in academic English courses to prepare them for dissertation writing and academic publishing.

To gain insights into SAI, we adopted a purposeful sampling to understand the differences in SAI patterns, their correlation with AI literacy levels, and how these affected writing performance. The study included 36 Chinese students in the MSc Digital Education program. These students were all in their first year and included three males and 33 females aged 21 to 30. Before the experiment, all participants were asked to answer AI literacy level questionnaires (7-point Likert-scale) adapted from Long and Magerko (2020) that include, for instance, 'I can distinguish between devices utilizing AI and devices that are not.', 'I can tell the difference between strong and weak AI.', and 'I can explain how to collect, analyze, interpret, and utilize data necessary for AI development.' The median score (5.09) of the AI literacy questionnaires were used to assign students into either a high (>5.09) or low group (≤ 5.09). This study received ethical approval from the [corresponding author's institute]'s Institutional Review Board (IRB no. NO. ER-AOFE-1278116462022090323481 2) and informed consent from all participants.

GenAI writing system

The research team developed the GenAI writing system by utilizing Unity (version 2021.3.17f1) as presented in Fig. 1.

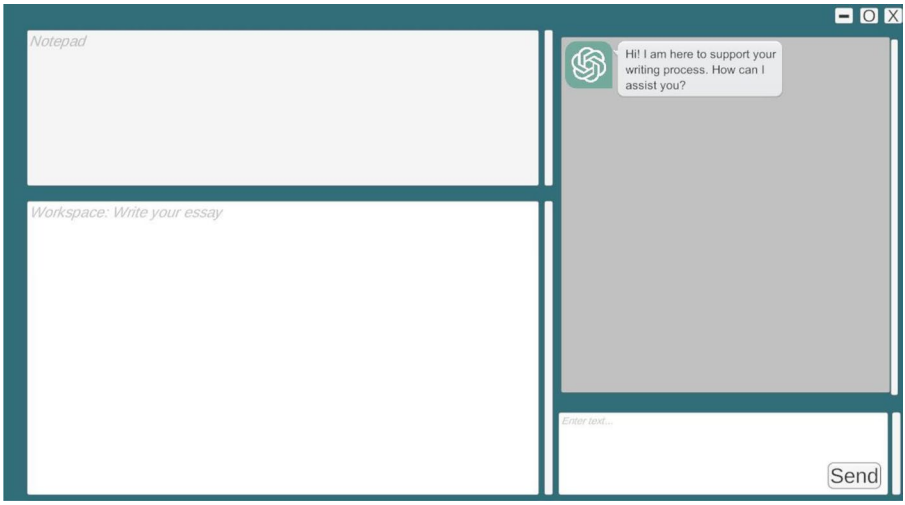

Fig. 1 The interface of the GenAI writing system used in the experiment

The system was designed to operate in a standalone mode on the Windows platform. The AI component employed the "gpt-4-0613" iteration of the Generative Pre-trained Transformer 4 (GPT-4) as provided by OpenAI. The user interface consists of four main components: the chat box, the message input box, the workspace, and the notepad. Situated in the upper right corner, the chat box employs a design similar to traditional instant messaging systems. This allows for an intuitive and familiar interface for the user to interact with the AI through text-based communication. The message input box, located in the lower right corner, serves as the primary input field for this interaction. The workspace, occupying the lower left corner, is designed to write the essay. This feature provides users with a space for continuous and uninterrupted writing. Lastly, the notepad, positioned in the upper left corner, serves as a reflection amplifier (Verpoorten et al., 2017), functioning not only as a tool for drafting and note-taking but also promotes deeper cognitive and metacognitive reflection. This allows users to write down ideas, organize thoughts, or draft sections of their text before finalizing their composition in the workspace. This four-part design creates a comprehensive and efficient user interface to meet the various aspects of the writing process. The chat history and other necessary data (e.g., the timestamp of each input action) are recorded and stored as a JavaScript Object Notation (JSON) file for analysis. The system was set up to record audio from users (e.g., students' think-aloud) and the screen of each individual's interaction on the system during the writing task operation.

Research procedure

To examine the SAI patterns on learning tasks, students were asked to think-aloud, verbalizing their thoughts and actions in their first language (Chinese), during the task in a one-on-one experiment setting (Kim et al., 2024). Before commencing the writing task, each participant watched the recorded video lecture on the use of GenAI

prompt skills for learning tasks and the think-aloud method for 45 min, practiced the think-aloud techniques, and explored the system guided by the research assistant. During the pilot experiments, students initially faced difficulties in verbalizing their thoughts in English, as they were more accustomed to writing quietly in their prior learning experiences. However, when allowed to verbalize their thoughts in Chinese while writing in English, students found it easier to complete the task, and following pre-experiment training, they encountered minimal challenges during the formal experiment in switching between languages. Then, the SAI on writing activity was conducted on the writing system (Fig. 1) described in the aforementioned section in the university computer lab via iMac (Retina 5 K, 27-inch, 2017). All students were randomly assigned between two topics closely related to their coursework (see Appendix 1). Although the task completion time differs among the students, most participants completed the writing activity within 1.5 h. The research team did not intervene while participants performed the task, except when their think-aloud paused for over 3 min. Following the method used in previous studies (e.g., Kim et al., 2024; Yoon et al., 2021), each participant's think-aloud was audio-recorded and each screen was video-recorded to capture their behavior which could be classified into conceptually meaningful categories representing SAI, such as reading and questioning GenAI's responses, copying and pasting AI's suggestions, or re-writing sentences, within the system while completing the writing tasks. The transcribed think-aloud protocol was translated by a professional translator (Chinese to English) who majored in both English translation and education to accurately convey participants' meanings between languages, and it was reviewed by the Chinese American faculty of Education major. This comprehensive setup allowed for a detailed examination of the students' interactions with the AI system and their writing processes.

Data collection and instruments

The key data collected for this study was participants' transcribed think-aloud protocol and the video-recorded behaviors. Consequently, the data were analyzed to probe into SAI patterns in academic writing using a coding scheme adapted from (Kim et al., 2024). Table 1 shows the three dimensions, subcategories, definitions, and exemplary discourse. We employed Excel to collect and sort out the students' think-aloud protocol. First, the first and third authors coded (dummy) all the think-aloud discourse data independently using the coding scheme in Table 1. We coded the entire think-aloud protocols and their behaviors (e.g., reading and questioning GenAI's responses, copying and pasting AI's suggestions, or re-writing sentences) from the recorded video, representing students' miscellaneous reactions or behaviors shown at least once for every activity to GenAI's responses, some of which occurred within 1 s during the task. To ensure the reliability of the coding results, the Cohen's Kappa Score of the two coders was calculated (0.82), where all disagreements were fully negotiated and resolved through discussion.

Table 1 Coding scheme

Dimension	Sub-category	Definition	Example
Writing activity (WA)	Ideation	Activating prior content knowledge or experience to build on ideas for the given writing topic.	Let me think about how I felt when I was in school. I did a lot of fun activities with a laptop during the class.
	Drafting	Writing a rough draft in response to the given writing topic.	Let me first write a draft to organize my thoughts and identify areas for improvement or questions I can ask AI.
	IA1	Inviting GenAI to build upon, elaborate, clarify, comment on, or improve the student's own ideas.	This is what I think. Can you review it and make any suggestions to further improve it?
	IA2	Simply adopting GenAI's suggestions without further elaboration or critique.	Let me copy this.
	IA3	Repeatedly modifying and refining the prompt or information given to GenAI to enhance GenAI's understanding of the student's intention.	If I give some additional explanation and examples for this, AI can work better or know what it means here.
	IA4	Reasoning or critiquing GenAI's suggestions.	Hey AI, what are you talking about? Are you following the context correctly?
	IA5	Integrating GenAI's recommendations for revision into the student's writing, with modification to the recommendations.	I think AI is making a point too, but not a point for young Chinese learners. Let me contextualize its suggestion and synthesize it with my thoughts.
Meta-cognitive activity (MC)	Planning	Developing a strategic plan to support writing task performance, such as defining the available resources and limitations to complete the task, sequencing activities, and distributing roles.	Big ideas should be determined first. So, let me first brainstorm to develop the main arguments with AI and then work on finding evidence.
	Orchestration	Adjusting the conceptual structure, argumentation, or direction of the written response by questioning, disagreeing with, or challenging an idea.	AI is giving me information on how AI can significantly influence various aspects of young learners' development, including cognitive skills and even socio-emotional aspects. Well, let me then shift the argumentation focus from the ethical challenge of AI in education to AI for holistic learning.
	Evaluation	Evaluating the task outcomes and the task completion process.	The first two arguments sound strong, but the last one lacks evidence. I should ask AI to find relevant statistics or data to support this claim.

Table 1 (continued)

Dimension	Sub-category	Definition	Example
Socio-emotional activity (SE)	PE	Expressing feelings of emotional closeness, personal association, appreciation, and affective connection with GenAI.	Thank you, AI! That was a clever idea.
	NE	Expressing negative emotions, including anger, anxiety, fear, disgust, disappointment, shame and guilt, antipathy, and hatred.	How many times I should repeat this and explain it with different examples? You are an idiot!

Analysis for student-generative AI interaction difference

First, the Chi-square test was first employed to discern the frequency differences in SAI codes to answer (H1) the SAI differences between different levels of AI literacy (high/low) groups. The frequency distribution allows us to grasp the big picture of SAI activities between the two groups; for instance, if the total number of 'ideation' and 'planning' activities of A group is higher from B group, it shows how much effort the A group put in before actual activity (writing) participation with Gen AI. Second, the epistemic network analysis (ENA) was conducted to investigate (H2) whether and what differences during SAI process occurred in completing a writing task between different levels of AI literacy (high/low) groups. The ENA model computes the co-occurrence of cognitive elements to reveal their interactions by incorporating units (the objects of the analysis, such as gender groups), stanzas (the scope of co-occurrence of elements), and codes (conceptual elements). Constructing an adjacencies matrix of the co-occurrence data, ENA subsequently illustrates the interconnection between cognitive elements within a two-dimensional space via normalization, reducing dimensions, and employing singular value decomposition (Swiecki et al., 2020). In ENA visualization, nodes portray the given codes (cognitive elements), and the thickness of the edges depicts the frequency of co-occurrence of the given codes (cognitive elements).

Since ENA can be an effective method to examine collaborative problem-solving processes (e.g., Swiecki et al., 2020), this study used the ENA web tool (http://app.epistemicnetwork.org) to investigate the encoded SAI data based on the coding scheme. The (high/low) AI literacy groups were set as the units of the ENA, and the stanza (the scope of co-occurrence of elements) was set to four. For the conceptual elements, 12 codes (sub-categories) from the coding scheme (see Table 1) were utilized to distinguish the different SAI patterns between the two groups. However, as the interaction between student and GenAI can be exhibited in multiple ways, this study divided the activity initiator and receiver (i.e., (a) student (alone), (b) student to GenAI, (c) Gen AI to student) to unravel who initiates the interaction and how much they interact. Then, a two-sample Mann-Whitney U test was employed to compare projection point distributions of high and low AI literacy groups as their distributions did not show normality (e.g., Swiecki et al., 2020) and to examine statistical significance.

Analysis for writing performance difference

To quantitatively assess the writing task performance differences between the AI literacy groups (H3), this study conducted expert evaluations based on the evaluation rubric. First, the evaluation rubric was developed based on the literature review (see Table 2) that includes three dimensions: content, structure/organization, and expression (Ahmad, 2019; Nguyen et al., 2024). Then, the evaluation rubric and the final essays via SAI were sent to five experts (including two faculty in literacy education, two faculty in teaching English to speakers of other languages, and one faculty in instructional design and technology) to score them using a 5-point Likert scale. All the information regarding participants was sealed for fair performance assessment. To investigate the performance difference between the AI literacy groups, the Wilcoxon rank-sum (Mann-Whitney U) test was conducted as the task performance scores did not show normality.

Results

Frequency distribution of student-generative AI interaction

As summarized in Table 3, the frequencies of the code distributions of writing activity and socio-emotional activity show statistical differences ($x^2=51.22$, $p\leq.001$; $x^2=5.62$, $p<.05$, respectively) while meta-cognitive activity did not ($x^2=0.83$, $p>.05$). In light of the frequency proportion, all activities seemingly demonstrate higher percentages of low AI literacy group (LL). However, the frequencies of most activities (except for Ideation, IA4, IA5, and orchestration) in the higher AI literacy group (HL) at least doubled.

Table 2 Evaluation rubric for students' academic essay

Category	Evaluation criteria	Score				
		1	2	3	4	5
Content (Focus, Purpose, controlling idea)	Is the essay supported with consistent evidence with originality and depth of ideas?					
	Is the essay presenting a fully developed position in answer to the question with relevant, fully extended, and well-supported ideas with valid and specific evidence?					
	Do ideas work together as a unified whole?					
Structure/organization	Is the essay organization sequential and appropriate to the topic?					
	Are paragraphs well developed and appropriately divided?					
	Are ideas linked with smooth and effective transition?					
Expression (Vocabulary choice, Grammatical range, and accuracy, Punctuation)	Does the essay use a wide range of vocabulary with very natural and sophisticated control of lexical features?					
	Does the essay have good control of errors in spelling, punctuation, and capitalization?					
	Is the essay structured effectively with a rich and well-chosen variety of sentence styles and length without grammar errors?					

Table 3 Frequency distribution of student-generative AI interaction result

Cat.	Code	High AI literacy group		Low AI literacy group		x^2	*p*-value
		freq	%	freq	%		
WA	Ideation	78	16.60%	69	30.26%	51.219	0.000
	Drafting	162	34.47%	81	35.53%		
	IA1	78	16.60%	12	5.26%		
	IA2	74	15.74%	12	5.26%		
	IA3	48	10.21%	26	11.40%		
	IA4	2	0.43%	5	2.19%		
	IA5	28	5.96%	23	10.09%		
	Total	470	43.56%	228	45.51%		
MC	Planning	173	36.42%	74	33.04%	0.833	0.659
	Orchestration	5	1.05%	3	1.34%		
	Evaluation	297	62.53%	147	65.63%		
	Total	475	44.02%	224	44.71%		
SE	PE	73	54.48%	17	34.69%	5.619	0.018
	NE	61	45.52%	32	65.31%		
	Total	134	12.42%	49	9.78%		

In light of the writing activity, HL is more likely to invite (IA1 = 78 times), adopt (IA2 = 74 times) Gen AI, and repeatedly modify statements (IA3 = 48 times) to involve it to collaborate in completing the writing task. In socio-emotional activity, HL expressed both positive (PE = 73 times) and negative (NE = 61 times) much more than LL (17 and 31 times, respectively). Although meta-cognitive activity did not show statistical differences between the two groups, the frequencies of overall meta-cognitive codes of HL (475 times) doubled LL (224 times). Planning (HL = 173 times; LL = 74 times) and Evaluation (HL = 297 times; LL = 147 times), in particular, present big differences in numbers, illustrating metacognitive activity differences between the two groups.

Epistemic network analysis (ENA)

The left side of Fig. 2 demonstrates overall ENA networks of SAI patterns on the writing project with the projected points (dots) and their centroids (squares) according to each group. It distinguishes the HL and LL groups in blue and red. The detailed SAI patterns of the two groups are discerned in the secondary plots according to their assigned colors.

The projected distribution points between the HL and LL groups in the ENA space were compared by deploying the Mann-Whitney *U* test (Swiecki et al., 2020), and the statistical group difference was found on the *x*-axis (ENA 1; Mdn=-0.14, $N = 10$ $U = 92.00$, $p = .000$, $r = -.84$), where it portrays [Student] and [GPT.Student] on the left and [Student.GPT] on the right. However, no statistically significant difference on the y-axis (ENA 2; Mdn=-0.10, $N = 10$ $U = 51.00$, $p = .97$, $r = -.02$) was found. Figure 3 shows the enlarged ENA results of HL and LL.

As summarized in Table 4, ENA connection coefficients of HL and LL groups show different SAI patterns discerned by the initiator. The (b) GenAI-student illustrates what activity GenAI engaged in to interact with a student, revealing that GenAI

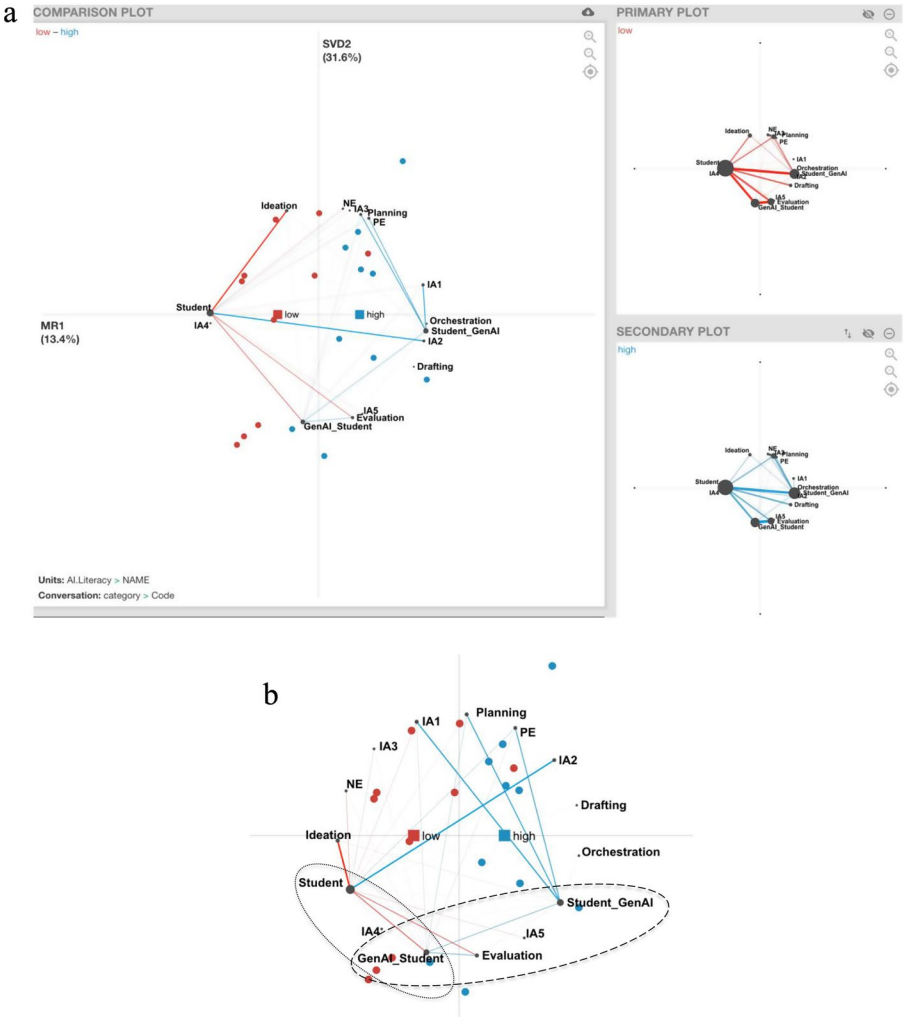

Fig. 2 **a**. Epistemic network analysis (ENA) result. **b**. Epistemic network analysis (ENA) result for visualization enhancement (equally spaced Fig. 2**a**.). *Note: The red indicates low AI literacy group, and the blue indicates high AI literacy group

is mainly involved in Evaluation (HL = 0.46, LL = 0.42) no matter the AI literacy level groups. The (a) student (alone) column indicates what kind of activity they participated in by themselves. It shows that students in the LL group distinctively ideate independently (Ideation = 0.23), whereas those in the HL group actively adopt what GenAI suggested (IA2 = 0.13). Along with this, the (c) student-GenAI that explains what SAI activity students commit to interacting with GenAI in completing the given task revealed that when planning the writing task, the HL group sent more comments and ideas to GenAI (Planning, HL = 0.28, LL. 0.21) with much higher positive emotion toward GenAI than the LL group (PE, HL = 0.12, Ll = 0.05).

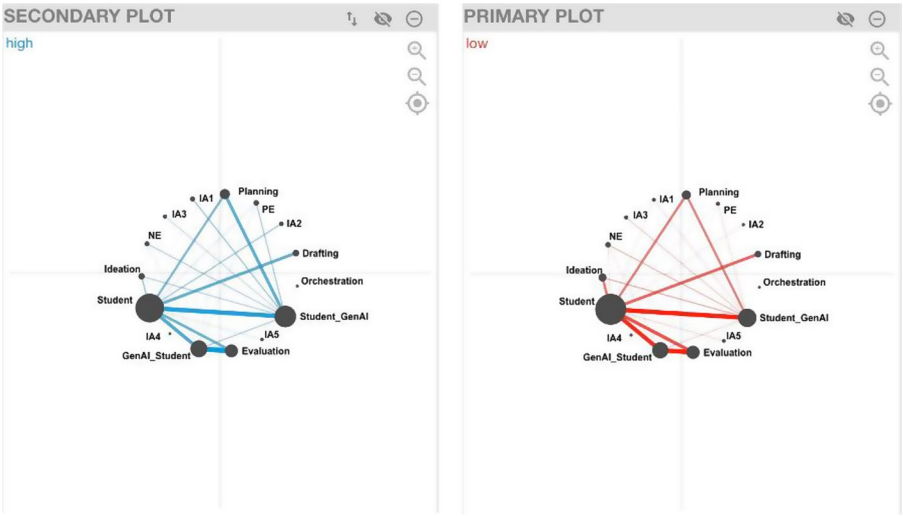

Fig. 3 AI literacy group difference (Epidemic network analysis) result. *Note: The high AI literacy group is on the left (blue), and the low group is on the right (red)

Table 4 ENA connection coefficients of HL and LL

Cat	Connection	(a) student (alone)		(b) GenAI-student		(c) student- GenAI	
		HL	LL	HL	LL	HL	LL
WA	Ideation	0.14	0.23	0.01	0.02	0.11	0.12
	Drafting	0.27	0.27	-	-	-	-
	IA1	0.01	0.03	0.02	-	0.13	0.05
	IA2	0.13	0.04	0.00	0.00	-	-
	IA3	0.02	-	-	0.01	0.08	0.08
	IA4	0.00	0.01	0.00	0.01	0.00	0.01
	IA5	0.05	0.07	0.00	0.00	-	-
MC	Planning	0.21	0.22	0.03	0.01	0.28	0.21
	Orchestration	0.01	0.01	0.00	-	0.01	-
	Evaluation	0.25	0.31	0.46	0.42	0.01	0.02
SE	PE	0.04	0.02	0.03	0.02	0.12	0.05
	NE	0.01	0.04	0.03	0.04	0.11	0.12

Writing performance (student-generative AI interaction effects)

Wilcoxon rank-sum (Mann-Whitney *U*) analyses were conducted to validate participant groups' writing performance differences regarding AI literacy level difference (high/low). As summarized in Table 5, the HL group received a statistically higher average total score (score difference = 3.40, p = .000) for the writing task via interacting with generative AI. Specifically, the HL group received significantly higher score points (score difference = 1.15, 1.05, 1.20, p = .000, respectively) for content, structure, and expression writing performance criteria than the LC group.

Table 5 Writing performance difference result

Category	AI literacy level (Mdn)		W	U	z
	High group	Low group			
Content	4.13	3.00	100	3	-3.24***
Structure	4.13	3.13	97	3	-3.24***
Expression	4.25	2.75	97	0.5	-3.42***
Total score	12.75	7.25	99.5	0	-3.45***

***$p \leq .001$, **$p < .01$, *$p < .05$

Discussion

The study found a difference in the frequency distribution of SAI between the groups during the writing task process. It is worth noting that the HL group demonstrated higher frequencies of performing in every category of activity (see Table 3). In particular, while the LL group participated more on IA4, reasoning or critiquing GenAI's suggestions, compared to the HL groups (HL = 2, LL = 5), the HL group had higher frequencies of performing the rest of IA-related activities compared to the LL group (IA1, HL = 78, LL = 12; IA2, HL = 74, LL = 12; IA3, HL = 48, LL = 26; IA5, HL = 28, LL = 23). Along with this, the HL group utilized GenAI in MC-related activities, such as planning (HL = 173, LL = 74), orchestration (HL = 5, LL = 3), evaluation (HL = 297, LL = 147). It is even more clearly evident through ENA analysis of the groups' SAI pattern differences. Similar to Kim et al. (2025), showcasing that the SAI of the LL group was identified as student-driven, less interactive, and jointly coordinated with GenAI (see dotted circle in Fig. 3b), the LL group tends to perform the writing tasks independently, rather than interacting with AI, with the students ideating for the given writing topic, evaluating the task outcomes, and assessing the completion process themselves. In contrast, the HL group actively accepts GenAI's suggestions, involves GenAI in planning, and allows GenAI to evaluate their work process and the task outcomes (see dashed circle in Fig. 3b). These findings indicate that the HL group demonstrates collaborative skills with AI, actively spending time to understand the capabilities of AI, including both its strengths and challenges. The HL group hone their perspectives and knowledge to gain valuable insights from GenAI' suggestions, recognizing the value of engaging GenAI in various decision-making processes of the writing task (i.e., planning, orchestration of the writing structure and contents, evaluating the task outcomes, and completion process). HL students determine when it is appropriate to delegate the tasks to GenAI and how to utilize and leverage the different or conflicting viewpoints presented by GenAI. In this regard, instructors should develop students' cognitive flexibility, an essential competency for adapting to new learning environments, including GenAI-assisted learning, for structuring and restructuring knowledge collaboratively with GenAI, and for understanding and solving unfamiliar problems with AI (OECD, 2013; Spiro et al., 2008). To do so, instructors may discern the pedagogical approaches that not only integrate GenAI in learning but do so in a way to guide students to expose collaborative learning experiences with AI. For instance, teachers may design and conduct student-AI collaborative reasoning, collaborative knowledge building, and problem-based learning by providing students with open and complex problems that could lead students to compare and contrast different approaches, alternative viewpoints, and contradictory data provided by GenAI to support consideration of a variety of solutions they may

not have otherwise considered on their own (Barak & Levenberg, 2016; Kim, 2024a). Furthermore, instructors can showcase and/or guide students to develop scenarios for varied roles, responsibilities, and contexts that can occur in SAI on learning tasks so that students can better consider new possibilities of effective SAI in learning (Kim & Cho, 2023; Kim, 2024b).

The study also found larger patterns in the socio-emotional relationship-related activities during SAI. Particularly, the HL group exhibited a positive emotional response to GenAI (HL = 0.12; LL = 0.05, see Table 4). This finding echoes prior research highlighting the value of socio-emotional interactions on task-related metacognition, involvement, and subsequent on-task motivation (Pekrun, 2005). However, it is important to note that simply experiencing joy-related emotions is insufficient to trigger interest in learning and engagement in productive collaborative learning. In this regard, research on student-student collaboration highlights the importance of emotion regulation, an ability to actively identify and understand one's emotions, interpret emotions in peer interactions in a more constructive manner, and maintain and regulate them into more positive emotions to fuel the learning process (Pekrun, 2005; Mänty et al., 2020). To account for socio-emotional interactions with students, GenAI, which aims to interact with students on learning tasks, should be grounded in the principles of emotional intelligence, emotional resilience, and psychological aspects of student learning, rather than simply fostering academic excellence and task performance. For instance, if students exhibit low activeness in the task or express negative emotional responses, the GenAI system can capture students' sentiments by analyzing varied data sources (i.e., chat history between student and AI, students' behavior). It should be proactive to guide students to be aware of their emotional state, appropriately manage their emotions and behavior more effectively, and develop self-management skills to complete learning tasks successfully.

Similarly, Meyer et al. (2024) reported that the automated, evidence-based feedback received during GenAI-assisted argumentative writing tasks induces achievement emotions such as enjoyment and pride, enhancing students' emotional experience and perseverance with a task in a GenAI-assisted learning environment. This gives them reasons to stay motivated and to actively keep working on the task at hand. To provide students with the necessary scaffolding to experience these positive effects of GenAI-assisted learning environments, instructors could inform students of both the strengths and weaknesses of AI in performing learning tasks so that students can identify opportunities and anticipated challenges and develop strategies for effectively working with AI (Kim et al., 2024). Yet, the findings of this study pose new questions about what makes the regulation of emotions effective within an SAI context. Future studies could further explore the types and sources of positive emotions students experience in SAI on learning tasks, more detailed socio-emotional regulation processes during SAI, and socio-emotional regulation strategies between a student and a GenAI system.

Conclusion

This study analyzed SAI patterns based on students' level of AI literacy and examined the difference in AI-assisted academic writing performance between the groups. To this end, the study has presented empirical evidence on the SAI process, the relationship between SAI patterns and GenAI-assisted writing task performance of students depending on their level of AI literacy. The study findings could contribute to theory, knowledge, and AIED practice by providing a foundation for a model of the SAI patterns, a predictor of SAI performance, and offer implications for designing educational AI and AIED instruction.

However, there are some limitations that future research should take into consideration. First, the data is from a single course with only 36 students in a graduate school performing a writing task. Future studies could extend the key findings of this study by conducting research in multiple course settings with various learning tasks and expanding the sample size. Additionally, although the level of AI literacy is a crucial student factor that influences the SAI, other characteristics such as gender, prior knowledge of and experience with GenAI, academic discipline, attitude toward AI, and so on would inevitably affect the SAI patterns. It is important to recognize that students' improved performance in AI-assisted writing tasks may also significantly depend on their foundational academic skills and general literacy, which precede and underpin their ability to leverage AI tools effectively. For instance, students with robust general literacy skills—such as critical thinking, comprehension, and advanced language use—may naturally excel in integrating AI feedback due to their baseline capabilities, independent of their AI literacy. This perspective suggests that the relationship between AI literacy and writing performance might be more complex and multifaceted. In light of this, we emphasize that while AI literacy remains a significant factor influencing student-AI interaction outcomes, it operates alongside a spectrum of underlying academic abilities. Future research should disentangle these variables to better understand the interplay between general literacy, AI literacy, and writing performance. We propose that subsequent studies consider a broader range of individual traits, including baseline academic skills, to paint a more comprehensive picture of the factors influencing success in AI-augmented learning environments. Finally, although the ENA was conducted with care to ensure the validity, the analysis of discourse data is insufficient for an in-depth understanding of SAI. To address such issues, the integration and synchronization of multimodal physiological data streams such as eye-tracking and electroencephalography (EEG) data could be considered to assess sentiment and behavioral patterns during SAI with different AI literacy level groups.

[Appendix 1: writing topics]

Topic 1: Some schools call for a ban on digital devices such as mobile phones, laptops, and tablet PCs from school to tackle classroom disruption, improve learning, and help protect young learners from cyberbullying, while others suggest that bans

by themselves will NOT necessarily lead to more engaged learning. Discuss which side you agree with and give your opinion.

When arguing either for banning devices in the classroom or against Banning Devices, include a range of potential classroom policies/strategies and ways to manage the use of digital devices in the classroom as reflecting and applying your learning in EDS431.

Topic 2: Among key concepts and theories discussed in EDS431, pick ONE theory/concept and discuss its strengths and weaknesses (limitations/criticism). When discussing its weaknesses, include how to address them when applying them to the digital education curriculum design.

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Data availability The data supporting the findings of this study are available from the corresponding author upon reasonable request. We recognize the importance of transparency and reproducibility in scientific research, and therefore, we are committed to sharing our data with interested parties for further analysis and verification. However, we reserve the right to assess the reasonableness of each request based on factors such as the purpose of the request, the ability of the requester to handle sensitive data responsibly, and any potential conflicts of interest. We encourage the use of our data for non-commercial research purposes and will make every effort to accommodate reasonable requests in a timely manner.

Code availability N/A.

Declarations

Ethical approval This study received ethical approval from the university's Institutional Review Board (NO. ER-AOFE 1278116462022090323481 2).

Consent to participant This study received informed consent from all participantsd.

Consent for publication The author accepts responsibility for releasing this material to be published.

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